2 hours on examples

<u>Problem 2 (USAMO 1973/4)</u>

We claim $x = y = z = 1$ is the only solution which obviously works.

We let x, y, z be the roots of the polynomial,

$$P(X) = X^3 + a_2X^2 + a_1X + a_0$$

By Newton Sums with $a_3 = 1$,

$$0 = p_1 + a_2$$

$$\Leftrightarrow a_2 = -p_1 = -3;$$

$$0 = p_2 + p_1a_2 + 2a_1$$

$$\Leftrightarrow a_1 = \frac{-p_2 - p_1a_2}{2} = 3$$

$$0 = a_3 + p_2a_2 + p_1a_1 + 3a_0$$

$$\Leftrightarrow a_0 = -\frac{a_3 + p_2a_2 + p_1a_1}{3} = -\frac{3 - 9 + 9}{3} = -1$$

Therefore,

$$P(X) = X^3 - 3X^2 + 3X - 1$$

$$= (X - 1)^3$$

Therefore, all the roots are $x = y = z = 1$.

<u>Problem 4 (16HMMTGUTS27)</u>

By Newton Sums,

$$0 = a_np_2 + a_{n-1}p_1 + 2a_{n-2}$$

$$\Leftrightarrow p_2 = -(-4 \cdot (4) + 16) = 0$$

Therefore,

$$\sum_{cyc} \frac{r_1^2}{r_2^2 + r_3^2 + r_4^2} = \sum_{cyc} \frac{r_1^2}{-r_1^2} = -4$$

<u>Problem 7 (JASONMAO)</u>

By Vieta's,

$$r + s + t = 7 \Leftrightarrow r + s = 7 - t$$

We want $m \coloneqq 7 - t$ to be a root, so we may plug in,

$$t^3 - 7t^2 + 3t + 2 = 0$$

$$\Leftrightarrow (7-m)^3 - 7(7-m)^2 + 3(7-m) + 2 = 0$$

$$\Leftrightarrow (343 - 147m + 21m^2 - m^3) - (343 - 98m + 7m^2) + (21 - 3m) + 2 = 0$$

$$\Leftrightarrow -m^3 + 14m^2 - 52m + 23 = 0$$

Therefore, $f(x) = x^3 - 14x^2 + 52x - 23$ has $r + s$ as a root. We may get,

$$-14 + 52 - 23 = 15$$

7 clubs; 3.5 hours

<u>Problem 12 (BMCQ54)</u>

We let a, b, c, d be the roots of a fourth-degree monic polynomial $F(X) = (X - a) \dots (X - d)$. Since,

$$a + b + c + d = 0;$$

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} = 0 \Longrightarrow bcd + acd + abd + abc = 0$$

Therefore, by Vieta's the coefficient of the x^3 and x terms are 0. We may write,

$$F(X) = X^4 + mX^2 + n$$

Therefore, if x is a root, then $-x$ is also a root. Hence, roots for pairs that add to zero.

<u>Problem 15 (11SMTA7)</u>

We see that, for $x = 2^0, 2^1, \dots, 2^{2010}$,

$$P(x) + 1 = P(2x)$$

Therefore, the polynomial,

$$G(x) \coloneqq P(2x) - P(x) - 1$$

Has roots at $x = 2^0, 2^1, \dots, 2^{2010}$. Note that G has degree 2011 as well. Therefore, we may write,

$$G(x) = k(x - 2^0)(x - 2^1) \dots (x - 2^{2010})$$

Plugging in $G(0) = P(0) - P(0) - 1 = -1$, we find,

$$k = 2^{-1005 \cdot 2011}$$

By Vieta's, we'd then get the degree one coefficient is,

$$\frac{2^{1005\cdot 2011}\left(\frac{1}{2^0}+\frac{1}{2^1}+\cdots+\frac{1}{2^{2010}}\right)}{2^{1005\cdot 2011}} = 2-\frac{1}{2^{2010}}$$

Note that the degree 1 coefficient of G is the degree one coefficient of P, so we are done.

Problem 16 (NOTVIETA)

We claim $\frac{r}{s}+\frac{s}{t}+\frac{t}{r};\frac{s}{r}+\frac{t}{s}+\frac{r}{t}$ are the roots of a monic quadratic with rational coefficients.

By Vieta's on our quadratic, since the sum of the roots $\frac{r}{s}+\frac{s}{t}+\frac{t}{r}+\frac{s}{r}+\frac{t}{s}+\frac{r}{t}=\frac{(rt+sr+ts)(r+s+t)}{rst}-3$ is rational by Vieta's on the cubic, the coefficient of the x term is rational.

By Vieta's on our quadratic, since the sum of the roots $\left(\frac{r}{s}+\frac{s}{t}+\frac{t}{r}\right)\left(\frac{s}{r}+\frac{t}{s}+\frac{r}{t}\right)=\frac{(rt+sr+ts)(r+s+t)}{rst}$ is rational by Vieta's on the cubic, the constant is also rational.

By the quadratic formula, we are done.

Problem 18 (13SMTA9)

$$=-\frac{a^4(b-c)+b^4(c-a)+c^4(a-b)}{(a-b)(b-c)(c-a)}$$

$$=-\frac{a^4b-a^4c+b^4c-b^4a+c^4a-c^4b}{(a-b)(b-c)(c-a)}$$

$$=-\frac{(a-b)\left((b-c)a^3+(b^2-bc)a^2+(b^3-b^2c)a+(c^4-b^3c)\right)}{(a-b)(b-c)(c-a)}$$

$$=-\frac{(a-b)(b-c)(a^3+a^2b+ab^2-b^2c-bc^2-c^3)}{(a-b)(b-c)(c-a)}$$

$$=\frac{(a^3+a^2b+ab^2-b^2c-bc^2-c^3)}{(a-c)}$$

$$=a^2+ab+ac+b^2+bc+c^2$$

$$=\frac{(a+b)^2+(b+c)^2+(c+a)^2}{2}$$

$$=30$$

Problem 19 (Z5394300)

If n is even, we'd get from RHS that $a=b=c=d=0$ since they're all real.

We may do casework in the n is odd case.

Case 1: There are > 1 zeroes in $\{a, b, c, d\}$. We have a pair of zeroes that sum to zero.

Case 2: There is 1 zero in $\{a, b, c, d\}$.

WLOG, let $d = 0$.

$$a + b + c = a^n + b^n + c^n = 0$$

Since a, b, c are non-zero, WLOG, let one be negative and two positives (we can just negate both sides). WLOG, let c be the negative. We let, $C \coloneqq -c$. Therefore, all a, b, C are positive. We get,

$$a + b = C;$$

$$a^n + b^n = C^n$$

$$\Leftrightarrow a^n + b^n = (a + b)^n$$

Since a, b are positive and $n > 1$, this is impossible by binomial expansion.

Case 3: All a, b, c, d are non-zero.

Subcase a: Three positive, one negative. (Symmetric with three negatives, one positive)

WLOG let d be negative, let $D \coloneqq -d$. Therefore,

$$a + b + c = D;$$

$$a^n + b^n + c^n = D = (a + b + c)^n$$

Expanding, since a, b, c are positive and $n > 1$, this is impossible.

Subcase b: Two positives, two negatives

WLOG let c, d be negative. Let $D \coloneqq -d; C \coloneqq -c$. We'd get,

$$a + b = C + D;$$

We can let $s \coloneqq a + b = C + D$.

$$a^n + b^n = C^n + D^n$$

WLOG, let $a \geq b; C \geq D$. We claim that $a = C; b = D$.

$$\Leftrightarrow a^n - C^n = D^n - b^n$$

$$\Leftrightarrow (a - C)(a^{n-1} + a^{n-2}C + \cdots + C^n) = (D - b)(D^n + D^{n-1}b + \cdots + b^n)$$

Using $a - C = b - D$,

$$\Leftrightarrow (a - C)(a^{n-1} + a^{n-2}C + \cdots + C^n) = -(a - C)(D^n + D^{n-1}b + \cdots + b^n)$$

Since a, b, C, D are positive, this is only possible when $a = C$. We may then get $b = D$. Therefore, in this case, we'd still get $a + c = 0; b + d = 0$.

26 clubs; 6.5 hours

Problem 20 (12BLACKMOP)

We let,

$$x \coloneqq \sqrt{a+h_B}; y \coloneqq \sqrt{b+h_C}; z = \sqrt{c+h_A}$$

Similarly, let,

$$x' \coloneqq \sqrt{a+h_C}; y' \coloneqq \sqrt{b+h_A}; z' = \sqrt{c+h_B}$$

We may let these be the roots of the unique third-degree monic polynomials $F(x); F'(x)$.

Let,

$$F(x) = x^3 + rx^2 + sx + t;$$

$$F'(x) = x^3 + r'x^2 + s'x + t'$$

The problem statement gives us $r = r'$. We may also get,

$$x^2 + y^2 + z^2 = x'^2 + y'^2 + z'^2$$

$$\Leftrightarrow (x+y+z)^2 - 2(xy+yz+zx) = (x'+y'+z')^2 - 2(x'y'+y'z'+z'x')$$

$$\Leftrightarrow s = s'$$

We let A be the area of the triangle.

$$2A = ah_A = bh_B = ch_C$$

We see that,

$$\Leftrightarrow 2A(a+b+c+h_A+h_B+h_C) = 2A(a+b+c+h_A+h_B+h_C)$$

$$\Leftrightarrow abc + abh_A + ach_C + bch_B + ah_Ch_A + bh_Bh_A + ch_Bh_C + h_Ah_Bh_C$$

$$= abc + abh_B + ach_A + bch_C + ah_bh_A + bh_Bh_c + ch_ah_C + h_Ah_Bh_C$$

$$\Leftrightarrow (a+h_B)(b+h_C)(c+h_A) = (a+h_C)(b+h_A)(c+h_B)$$

$$\Leftrightarrow xyz = x'y'z'$$

Therefore, $t' = t$ as well.

We get that $F(x) = F'(x)$ for all x. Therefore, they also share roots.

$$\Leftrightarrow \sqrt{a+h_B}, \sqrt{b+h_C}, \sqrt{c+h_A} \text{ are } \sqrt{a+h_C}, \sqrt{b+h_A}, \sqrt{c+h_B} \text{ in some order,}$$

$$\Leftrightarrow a+h_B, b+h_C, c+h_A \text{ are } a+h_C, b+h_A, c+h_B \text{ in some order.}$$

For the sake of contradiction, let ABC be non-isosceles. WLOG, let $a > b > c$. Obviously,

$$a + h_B \neq a + h_C; a + h_B \neq c + h_B$$

Therefore, $a + h_B = b + h_A \Leftrightarrow \frac{ab+2A}{b} = \frac{ab+2A}{a} \Leftrightarrow a = b$. (!)

Problem 21 (H2954369)

We may get,

$$\frac{a}{b}+\frac{b}{c}+\frac{c}{d}+\frac{d}{e}+\frac{e}{a}=5+\frac{k}{2}$$

Subtracting from equations from this, we can solve to get,

$$\frac{a}{b}=-1+\frac{k}{2};$$

$$\frac{b}{c}=2-\frac{k}{2};$$

$$\frac{c}{d}=\frac{k}{2};$$

$$\frac{d}{e}=3-\frac{k}{2};$$

$$\frac{e}{a}=1+\frac{k}{2}$$

We see that, if $a=a_1, b=b_1, \dots, e=e_1$ is a solution, then we can scale each term and it is still a solution. Therefore, WLOG, let $a=1$.

Note that,

$$b=\frac{1}{-1+\frac{k}{2}}; c=\frac{b}{2-\frac{k}{2}}=\frac{1}{\left(-1+\frac{k}{2}\right)\left(2-\frac{k}{2}\right)}; \dots; e=\frac{d}{3-\frac{k}{2}}=\frac{1}{\left(-1+\frac{k}{2}\right)\left(2-\frac{k}{2}\right)\left(\frac{k}{2}\right)\left(3-\frac{k}{2}\right)}$$

Therefore, as long as none of the factors of $\left(-1+\frac{k}{2}\right)\left(2-\frac{k}{2}\right)\left(\frac{k}{2}\right)\left(3-\frac{k}{2}\right)\left(1+\frac{k}{2}\right)$ can be 0. Finally, the last equation yields,

$$1=\left(-1+\frac{k}{2}\right)\left(2-\frac{k}{2}\right)\left(\frac{k}{2}\right)\left(3-\frac{k}{2}\right)\left(1+\frac{k}{2}\right)$$

Note that any k that satisfies this solution already satisfies $0 \neq \left(-1+\frac{k}{2}\right)\left(2-\frac{k}{2}\right)\left(\frac{k}{2}\right)\left(3-\frac{k}{2}\right)\left(1+\frac{k}{2}\right)$. Therefore, we're looking for the sum of the fourth roots of,

$$f(k)=\left(-1+\frac{k}{2}\right)\left(2-\frac{k}{2}\right)\left(\frac{k}{2}\right)\left(3-\frac{k}{2}\right)\left(1+\frac{k}{2}\right)-1$$

This polynomial shares roots with,

$$g(k)=32f(k)=(k-6)(k-4)(k-2)(k)(k+2)-32$$

However, note that, by Newton sums, the constant term of a quintic does not affect p_1, p_2, p_3, p_4. Therefore, our answer is, $p_4=2^4+(-2)^4+4^4+6^4+0^4=1584$.

36 clubs; 8 hours

Problem 23 (Longlist 1985/19)

We let $a \coloneqq \sqrt{x}; b \coloneqq -\frac{1}{y}; c \coloneqq 2w; d = -3z$. Therefore, the given is,

$$a + b + (-1)^1 + (-1)^1 = (-2)^1 + 1^1 + c + d$$

$$a^2 + b^2 + (-1)^2 + (-1)^2 = (-2)^2 + 1^2 + c^2 + d^2;$$

$$a^3 + b^3 + (-1)^3 + (-1)^3 = (-2)^3 + 1^3 + c^3 + d^3;$$

$$a^4 + b^4 + (-1)^4 + (-1)^3 = (-2)^4 + 1^4 + c^4 + d^4$$

Let $f(x), g(x)$ be a monic quartic polynomial with roots respectively $(a, b, -1, -1)$ and $(-2, 1, c, d)$. By Newton Sums, these two polynomials are the same.

Therefore, $c = d = -1$ and a, b are $-2, 1$ in some order. Noting that $a \geq 0$, we get,

$$a = 1; b = -2; c = -1; d = -1$$

Therefore,

$$x = 1; y = \frac{1}{2}; w = -\frac{1}{2}; z = \frac{1}{3}$$

Problem 25 (HMMT 2020 A9)

We may write,

$$Q(\alpha^2)^2 = \frac{\prod_{i=1}^{2020} \prod_{j=1}^{2020} (a^2 - r_i r_j)}{\prod_{i=1}^{2020} (\alpha^2 - r_i^2)}$$

$$= \frac{\prod_{i=1}^{2020} r_i^{2020} \prod_{j=1}^{2020} \left(\frac{a^2}{r_i} - r_j\right)}{\prod_{i=1}^{2020} (\alpha^2 - r_i^2)}$$

$$= \frac{\prod_{i=1}^{2020} r_i^{2020} P\left(\frac{a^2}{r_i}\right)}{\prod_{i=1}^{2020} (\alpha^2 - r_i^2)}$$

$$= \frac{\prod_{i=1}^{2020} r_i^{2020} \left(\frac{a^{4040}}{r_i^{2020}} + \frac{a^2}{r_i} + 2\right)}{\prod_{i=1}^{2020} (\alpha^2 - r_i^2)}$$

$$= \frac{\prod_{i=1}^{2020} a^{4040} + a^2 r_i^{2019} + 2r_i^{2020}}{\prod_{i=1}^{2020} (\alpha^2 - r_i^2)}$$

Now note that, $a^{2020} + a + 2 = 4 \Leftrightarrow a^{4040} = a^2 - 4a + 4$. Also, $r_i^{2020} = -r_i - 2$.

$$= \frac{\prod_{i=1}^{2020} a^2 - 4a + 4 + \frac{a^2(-r_i - 2)}{r_i} - 2r_i - 4}{\prod_{i=1}^{2020}(\alpha^2 - r_i^2)}$$

$$= \frac{\prod_{i=1}^{2020} \frac{1}{r_i} \prod_{i=1}^{2020} -4ar_i - 2a^2 - 2r_i^2}{\prod_{i=1}^{2020}(\alpha^2 - r_i^2)}$$

$$= \frac{\frac{1}{\prod_{i=1}^{2020} r_i} \prod_{i=1}^{2020} -2(a + r_i)^2}{\prod_{i=1}^{2020}(a - r_i)(a + r_i)}$$

By Vieta's to get $\prod_{i=1}^{2020} r_i = 2$,

$$= 2^{2020} \cdot \frac{1}{2} \prod_{i=1}^{2020} \frac{(a + r_i)}{(a - r_i)}$$

$$= 2^{2019} \cdot \frac{P(-a)}{P(a)}$$

$$= 2^{2019} \cdot \frac{a^{2020} - a + 2}{4}$$

$$= 2^{2019} \cdot \left(1 - \frac{2a}{4}\right)$$

$$= 2^{2019} - 2^{2018}a$$

Summing over all roots of a, we'd get,

$$\sum 2^{2019} - 2^{2018}a = 2020 \cdot 2^{2019} - 2^{2018} \sum a = 2020 \cdot 2^{2019} = 505 \cdot 2^{2021}$$

54 clubs; 10 hours